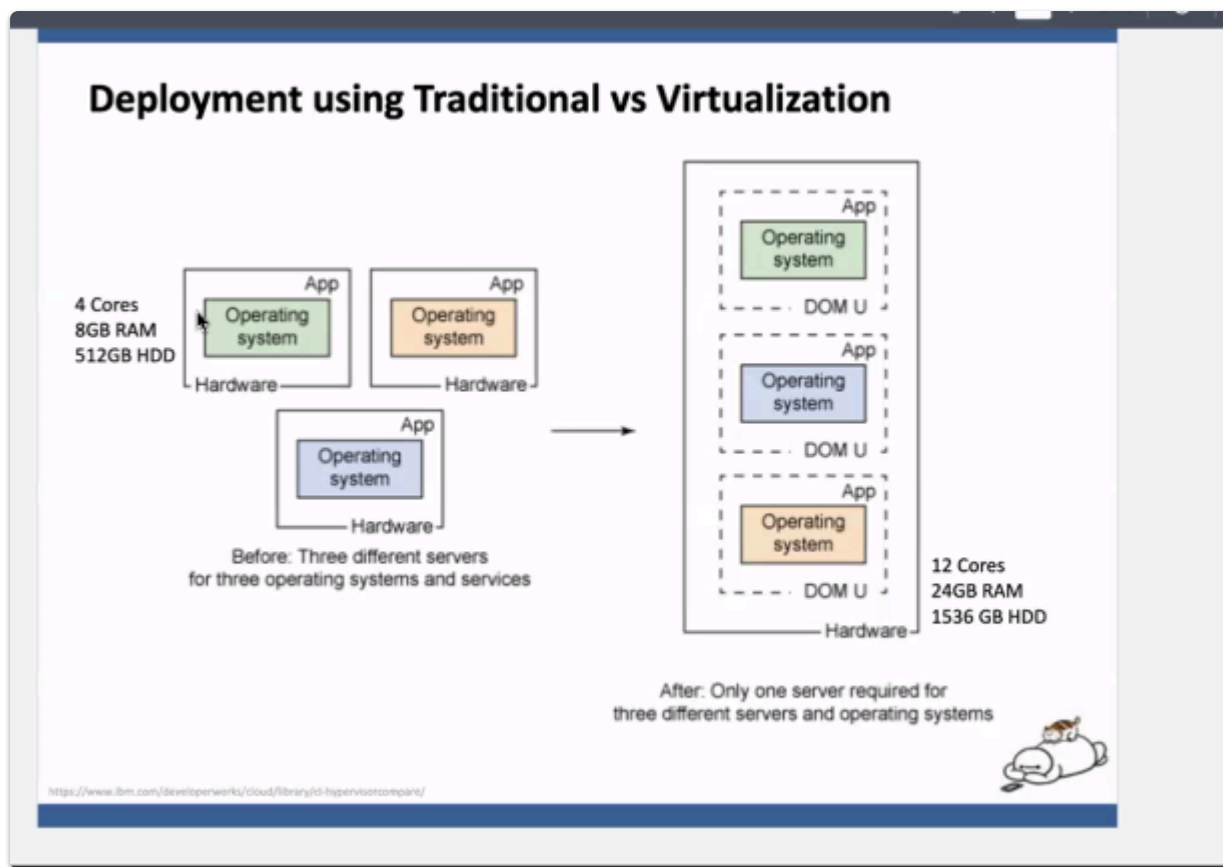


STCLOUD 05.19



Properties of Virtualization

1. Partitioning
 - divide resrouces between vms
 - run multiple os or vms in single physical amchione
2. Isolation
 - reduce risk of conflict in dependencies
3. Encapsulation
 - move and copy as files
 - improve portability
4. Hardware Independence
 - provision or migrate to any server without the need to perform reconfigs on vm

Server Virtualizoation

- Enable multi OS to run on a single physical server
- Partitions a physical server into multiple virtual servers

Benefits

- Greater IT efficiencies
- Reduced operating costs
- maximization of resource
- increased application performance since technically magisa sila. and kahit isolated sila, theyre still physically near each other
- higher server availability

Network Virtualization

- completeles reproducs virtual network allowing applications

Virtualization Food Groups

Essentially ano ano mga kinakain ng isang system

Desktop Virtualization

Deploying desktops as a managed service enables IT orgs to respond faster

Centralize

Types:

- Desktop as a Service DaaS
 - cloud service that deleivers virtualizaed destops
- Virtual Desktop infrastricture
 - deployed on an on p remise datacenter
- Local desktop cirtuasldation
 - hypervisor runs on local ocomputer conaitning vms
- Remote rdesktop service
 - remote access windows service

Storage Virtualization

- pulls physical storage from multiple sources to a single storage device
- can be pooled despite being heterogenous

Data Virtualization

- creates a software layer between data and applications

Virtualization Companies

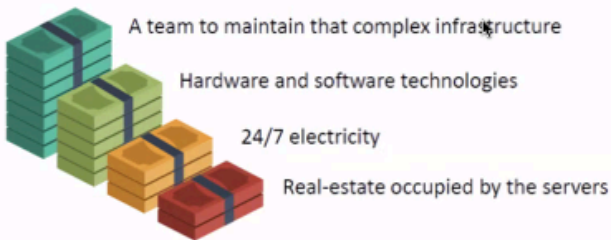


COMPANY	HYPERVERSOR	CLOUD TECHNOLOGY
VMware	ESXi	
Oracle	Oracle Virtualbox	OCI
Microsoft	Hyper-V	Azure
Citrix	Xen Server	Citrix Cloud Services
Redhat	KVM	Redhat Cloud Suite
Amazon		AWS
Google		Google Cloud Platform (GCP)

IaaS foundation

CHALLENGE # 1 - COST

Building and Running Private Data Centers



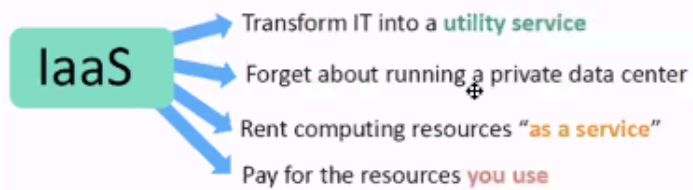
CHALLENGE # 1 - COST

Building and Running Private Data Centers



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IAAS BENEFITS

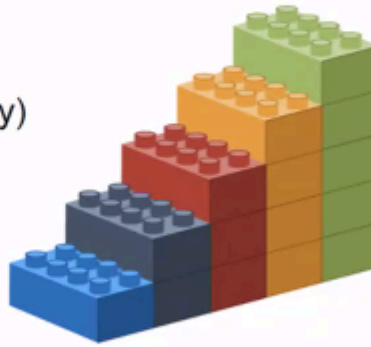


speaker notes

COMPUTE, STORAGE AND NETWORKING

IAAS Building Blocks

- Compute
- Storage
- Networking (and Security)
- Management



THE ARCHITECTURAL NEED

Amazon.com's e-commerce tools were "a jumbled mess:"

- Applications and architectures were built **without proper planning**
- Services had to be **separated** from each other

Solution: Tools became a set of well-documented APIs, which became the standard for service development at Amazon.

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THE ARCHITECTURAL NEED

Amazon.com still struggled to build applications quickly.

- Database, compute, and storage components took 3 months to build.
- Each team built their own resources, with no planning for scale or re-usability.

Solution: Built internal services to create highly available, scalable, and reliable architectures on top of their infrastructure. In 2006, started selling these services as AWS.

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AWS Regions

- AWS has **Regions** all around the world
- Names can be us-east-1, eu-west-3...
- A region is a **cluster of data centers**
- Most AWS services are **region-scoped**



US East (N. Virginia)	us-east-1
US East (Ohio)	us-east-2
US West (N. California)	us-west-1
US West (Oregon)	us-west-2
Africa (Cape Town)	
af-south-1	
Asia Pacific (Hong Kong)	
ap-east-1	
Asia Pacific (Mumbai)	ap-south-1
Asia Pacific (Seoul)	ap-northeast-2
Asia Pacific (Singapore)	ap-southeast-1
Asia Pacific (Sydney)	ap-southeast-2
Asia Pacific (Tokyo)	ap-northeast-1
Canada (Central)	
ca-central-1	
Europe (Frankfurt)	
eu-central-1	
Europe (Ireland)	eu-west-1
Europe (London)	eu-west-2
Europe (Paris)	eu-west-3
Europe (Stockholm)	eu-north-1
Middle East (Bahrain)	
me-south-1	
South America (São Paulo)	
sa-east-1	

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Week 2] STCLOUD - Introduction to Virtualization and Cloud Com... ☆

Slide Arrange Tools Extensions Help

7

ECS MODERNIZATION ARCHITECTURE

Service Regionalization will decentralize OCS into the country's main geographical areas and ensure chargeable services are still available for regions affected by a natural disaster or network disconnection

PRE-REGIONALIZATION

All rating events for Data wallets are currently managed by a single site located at NCR. If the natural disaster known as Big One occurs, network disconnection will happen thus data services will be unavailable on isolated regions

POST-REGIONALIZATION

During a natural disaster known such as the Big One which can result to network disconnection or region isolation, chargeable data services will still be available by routing the regional ISG and NTG nodes on the available ECS node to AWS.

Note: Additional AWS direct connect to be established on NLZ and VisMin Node

Active Regional Node

VisMin Node to be at 70% Capacity
NLZ Node at 30% HW Capacity

Background Layout Theme Transition

1 2 3 4 5 6 7 8 9

How to choose an AWS Region?

If you need to launch a new application, where should you do it?

- **Compliance** with data governance and legal requirements: data never leaves a region without your explicit permission
- **Proximity** to customers: reduced latency
- **Available services** within a Region: new services and new features aren't available in every Region
- **Pricing**: pricing varies region to region and is transparent in the service pricing page

add speaker notes

AWS Points of Presence (Edge Locations)

- Amazon has 400+ Points of Presence (400+ Edge Locations & 10+ Regional Caches) in 90+ cities across 40+ countries
- Content is delivered to end users with lower latency



<https://aws.amazon.com/cloudfront/features/>